

Jin-Chuan Shi

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Research Interests

My research lies at the intersection of **3D reconstruction**, **dexterous manipulation**, and **large-scale machine learning**. I work on reconstructing hand-object interaction from in-the-wild video and retargeting it into physically consistent simulation, turning Internet-scale human videos into robot-usable manipulation data for general-purpose dexterous manipulation.

Core research question: How can we build *scalable, efficient, and robust* reconstruction-and-retargeting systems that transform massive Internet videos into high-quality manipulation data — and, on top of such data, scalable robot learning systems that unify perception and dexterous manipulation, enabling reliable general-purpose robots in unstructured, real-world environments?

Education

Zhejiang University — College of Computer Science and Technology *Hangzhou, China*
Sept. 2025 – Mar. 2029 (expected)

Ph.D. in Computer Science and Technology • Advisors: Prof. Chunhua Shen, Prof. Hao Chen

Beihang University — School of Computer Science and Engineering *Beijing, China*
Sept. 2022 – Jan. 2025

M.Eng. in Computer Science and Technology • GPA 3.80/4.00 • Advisors: Prof. Shi-Min Hu (Tsinghua), Prof. Miao Wang

Beihang University — Shen Yuan Honors College *Beijing, China*
Sept. 2018 – June 2022

B.Eng. in Computer Science and Technology • GPA 3.73/4.00 • Advisor: Prof. Miao Wang

Publications

* Equal contribution. Author of this CV in **bold**.

- [1] **HORIZON: Recoverability-Governed Curriculum for Physical-Domain Scaling.**
Chenhao Bai, Liqin Lu, Kaijun Wang, Hui Chen, **Jin-Chuan Shi**, Yuyang Liu, Hao Chen, Chunhua Shen.
arXiv preprint, 2026.
- [2] **TriSplat: Simulation-Ready Feed-Forward 3D Scene Reconstruction.**
Weijie Wang*, Zimu Li*, **Jin-Chuan Shi**, Zeyu Zhang, Botao Ye, Marc Pollefeys, Donny Y. Chen, Bohan Zhuang.
arXiv preprint, 2026.
- [3] **AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation.**
Jin-Chuan Shi*, Binhong Ye*, Tao Liu, Xiaoyang Liu, Yangjinhui Xu, Junzhe He, Zeju Li, Hao Chen, Chunhua Shen.
ACM SIGGRAPH (Conference Track), 2026.
- [4] **AoE: Always-on Egocentric Human Video Collection for Embodied AI.**
Bowen Yang, Zishuo Li, Yang Sun, Changtao Miao, Yifan Yang, Man Luo, Xiaotong Yan, Feng Jiang, **Jin-Chuan Shi**, Yankai Fu, Ning Chen, Junkai Zhao, Pengwei Wang, Guocai Yao, Shanghang Zhang, Hao Chen, Zhe Li, Kai Zhu.
CVPR Workshop on Embodied AI in the Era of Large Models, 2026.
- [5] **Mono4DEditor: Text-Driven 4D Scene Editing from Monocular Video via Point-Level Localization of Language-Embedded Gaussians.**
Jin-Chuan Shi*, Chengye Su*, Jiajun Wang, Ariel Shamir, Miao Wang.
arXiv preprint, 2025.

- [6] **LEGaussians: Language Embedded 3D Gaussians for Open-Vocabulary Scene Understanding.**
Jin-Chuan Shi, Miao Wang, Hao-Bin Duan, Shao-Hua Guan.
IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR), 2024. *IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025.
- [7] **BakedAvatar: Baking Neural Fields for Real-Time Head Avatar Synthesis.**
Hao-Bin Duan, Miao Wang, **Jin-Chuan Shi**, Xu-Chuan Chen, Yan-Pei Cao.
ACM Trans. on Graphics (Proc. SIGGRAPH Asia), 2023.
- [8] **SceneFusion: Room-Scale Environmental Fusion for Efficient Traveling Between Separate Virtual Environments.**
Miao Wang, Yi-Jun Li, **Jin-Chuan Shi**, Frank Steinicke.
IEEE Trans. on Visualization and Computer Graphics (TVCG), 2023.
- [9] **Bullet Comments for 360° Video.**
Yi-Jun Li, **Jin-Chuan Shi**, Fang-Lue Zhang, Miao Wang.
IEEE Conf. on Virtual Reality and 3D User Interfaces (VR), 2022.
- [10] **Scene-Context-Aware Indoor Object Selection and Movement in VR.**
Miao Wang, Ziming Ye, **Jin-Chuan Shi**, Yong-Liang Yang.
IEEE Conf. on Virtual Reality and 3D User Interfaces (VR), 2021.

Honors & Awards

- National Scholarship, China Ministry of Education, 2024
- Outstanding Graduate Student, Beihang University, 2024
- Outstanding Graduates of Beijing's Higher Education Institutions, 2022
- Freshman Academic Scholarship (Special Prize), Beihang University, 2018
- Outstanding Academic Scholarship, Beihang University, 2019–2021, 2023
- Outstanding Student Leader Award, Beihang University, 2019–2021

Academic Service

Reviewer ACM SIGGRAPH (Asia), Pacific Graphics (PG), IEEE ISMAR, NeurIPS

Skills

Languages	Python, C, C++, CUDA, C#, Java, JavaScript, LaTeX
Tools	PyTorch, Unity3D, Linux, Git
Domains	3D Reconstruction, Neural Rendering, Gaussian Splatting, Hand-Object Interaction, Dexterous Manipulation, VR/AR
Spoken	English (fluent, CET-4 & CET-6), Mandarin (native)